



```
In [1]: import pandas as pd
```

```
In [2]: df= pd.read_csv("car_price.csv")
```

```
In [3]: df
```

```
Out[3]:
```

|      | car_name                                              | car_prices_in_rupee | kms_driven | fuel_type | transmission | own |
|------|-------------------------------------------------------|---------------------|------------|-----------|--------------|-----|
| 0    | Jeep<br>Compass<br>2.0<br>Longitude<br>Option<br>BSIV | 10.03 Lakh          | 86,226 kms | Diesel    | Manual       | 1st |
| 1    | Renault<br>Duster RXZ<br>Turbo CVT                    | 12.83 Lakh          | 13,248 kms | Petrol    | Automatic    | 1st |
| 2    | Toyota<br>Camry 2.5<br>G                              | 16.40 Lakh          | 60,343 kms | Petrol    | Automatic    | 1st |
| 3    | Honda Jazz<br>VX CVT                                  | 7.77 Lakh           | 26,696 kms | Petrol    | Automatic    | 1st |
| 4    | Volkswagen<br>Polo 1.2<br>MPI<br>Highline             | 5.15 Lakh           | 69,414 kms | Petrol    | Manual       | 1st |
| ...  | ...                                                   | ...                 | ...        | ...       | ...          | ... |
| 5507 | BMW X1<br>sDrive 20d<br>xLine                         | 28.90 Lakh          | 45,000 kms | Diesel    | Automatic    | 1st |
| 5508 | BMW M<br>Series M4<br>Coupe                           | 64.90 Lakh          | 29,000 kms | Petrol    | Automatic    | 2nd |
| 5509 | Jaguar XF<br>2.2 Litre<br>Luxury                      | 13.75 Lakh          | 90,000 kms | Diesel    | Automatic    | 2nd |
| 5510 | BMW 7<br>Series<br>730Ld                              | 29.90 Lakh          | 79,000 kms | Diesel    | Automatic    | 3rd |
| 5511 | BMW 5<br>Series<br>520d M<br>Sport                    | 31.90 Lakh          | 42,000 kms | Diesel    | Automatic    | 2nd |

5512 rows x 9 columns

```
In [4]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5512 entries, 0 to 5511
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   car_name               5512 non-null   object
1   car_prices_in_rupee    5512 non-null   object
2   kms_driven             5511 non-null   object
3   fuel_type              5510 non-null   object
4   transmission           5511 non-null   object
5   ownership              5509 non-null   object
6   manufacture            5507 non-null   float64
7   engine                 5511 non-null   object
8   Seats                  5511 non-null   object
dtypes: float64(1), object(8)
memory usage: 387.7+ KB

```

```
In [6]: df.isnull().sum()
```

```

Out[6]: car_name           0
        car_prices_in_rupee  0
        kms_driven         1
        fuel_type          2
        transmission       1
        ownership          3
        manufacture        5
        engine             1
        Seats              1
        dtype: int64

```

```
In [7]: df.dropna(inplace=True)
```

```
In [9]: df.isnull().sum()
```

```

Out[9]: car_name           0
        car_prices_in_rupee  0
        kms_driven         0
        fuel_type          0
        transmission       0
        ownership          0
        manufacture        0
        engine             0
        Seats              0
        dtype: int64

```

```
In [10]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 5500 entries, 0 to 5511
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   car_name              5500 non-null   object
1   car_prices_in_rupee  5500 non-null   object
2   kms_driven            5500 non-null   object
3   fuel_type            5500 non-null   object
4   transmission          5500 non-null   object
5   ownership             5500 non-null   object
6   manufacture          5500 non-null   float64
7   engine               5500 non-null   object
8   Seats                5500 non-null   object
dtypes: float64(1), object(8)
memory usage: 429.7+ KB
```

```
In [11]: df
```

Out[11]:

|             | car_name                                              | car_prices_in_rupee | kms_driven | fuel_type | transmission | own |
|-------------|-------------------------------------------------------|---------------------|------------|-----------|--------------|-----|
| <b>0</b>    | Jeep<br>Compass<br>2.0<br>Longitude<br>Option<br>BSIV | 10.03 Lakh          | 86,226 kms | Diesel    | Manual       | 1st |
| <b>1</b>    | Renault<br>Duster RXZ<br>Turbo CVT                    | 12.83 Lakh          | 13,248 kms | Petrol    | Automatic    | 1st |
| <b>2</b>    | Toyota<br>Camry 2.5<br>G                              | 16.40 Lakh          | 60,343 kms | Petrol    | Automatic    | 1st |
| <b>3</b>    | Honda Jazz<br>VX CVT                                  | 7.77 Lakh           | 26,696 kms | Petrol    | Automatic    | 1st |
| <b>5</b>    | Volkswagen                                            | 7.66 Lakh           | 49,719 kms | Petrol    | Automatic    | 1st |
| ...         | ...                                                   | ...                 | ...        | ...       | ...          | ... |
| <b>5507</b> | BMW X1<br>sDrive 20d<br>xLine                         | 28.90 Lakh          | 45,000 kms | Diesel    | Automatic    | 1st |
| <b>5508</b> | BMW M<br>Series M4<br>Coupe                           | 64.90 Lakh          | 29,000 kms | Petrol    | Automatic    | 2nd |
| <b>5509</b> | Jaguar XF<br>2.2 Litre<br>Luxury                      | 13.75 Lakh          | 90,000 kms | Diesel    | Automatic    | 2nd |
| <b>5510</b> | BMW 7<br>Series<br>730Ld                              | 29.90 Lakh          | 79,000 kms | Diesel    | Automatic    | 3rd |
| <b>5511</b> | BMW 5<br>Series<br>520d M<br>Sport                    | 31.90 Lakh          | 42,000 kms | Diesel    | Automatic    | 2nd |

5500 rows × 9 columns

```
In [13]: df["manufacture"]=df["manufacture"].astype("int")
```

```
In [14]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 5500 entries, 0 to 5511
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   car_name               5500 non-null   object
1   car_prices_in_rupee    5500 non-null   object
2   kms_driven             5500 non-null   object
3   fuel_type              5500 non-null   object
4   transmission           5500 non-null   object
5   ownership              5500 non-null   object
6   manufacture            5500 non-null   int64
7   engine                 5500 non-null   object
8   Seats                  5500 non-null   object
dtypes: int64(1), object(8)
memory usage: 429.7+ KB
```

```
In [15]: df["car_prices_in_rupee"].unique()
```

```
Out[15]: array(['10.03 Lakh', '12.83 Lakh', '16.40 Lakh', ..., '28.90 Lakh',
               '64.90 Lakh', '31.90 Lakh'], shape=(1298,), dtype=object)
```

```
In [ ]: df["price is Lakh"]=df["car_prices_in_rupee"].str.replace(" Lakh", "")
```

```
In [25]: df["price is Lakh"].astype("float")
```

```
Out[25]: 0      10.03
1      12.83
2      16.40
3       7.77
5       7.66
...
5507    28.90
5508    64.90
5509    13.75
5510    29.90
5511    31.90
Name: price is Lakh, Length: 5500, dtype: float64
```

```
In [22]: df["price is Lakh"]=df["price is Lakh"].str.replace(" Crore", "")
```

```
In [24]: df["price is Lakh"]=df["price is Lakh"].str.replace(",","")
```

```
In [26]: df["price is Lakh"]= df["price is Lakh"].astype(float)
```

```
In [27]: df
```

Out[27]:

|      | car_name                                              | car_prices_in_rupee | kms_driven | fuel_type | transmission | own |
|------|-------------------------------------------------------|---------------------|------------|-----------|--------------|-----|
| 0    | Jeep<br>Compass<br>2.0<br>Longitude<br>Option<br>BSIV | 10.03 Lakh          | 86,226 kms | Diesel    | Manual       | 1st |
| 1    | Renault<br>Duster RXZ<br>Turbo CVT                    | 12.83 Lakh          | 13,248 kms | Petrol    | Automatic    | 1st |
| 2    | Toyota<br>Camry 2.5<br>G                              | 16.40 Lakh          | 60,343 kms | Petrol    | Automatic    | 1st |
| 3    | Honda Jazz<br>VX CVT                                  | 7.77 Lakh           | 26,696 kms | Petrol    | Automatic    | 1st |
| 5    | Volkswagen                                            | 7.66 Lakh           | 49,719 kms | Petrol    | Automatic    | 1st |
| ...  | ...                                                   | ...                 | ...        | ...       | ...          | ... |
| 5507 | BMW X1<br>sDrive 20d<br>xLine                         | 28.90 Lakh          | 45,000 kms | Diesel    | Automatic    | 1st |
| 5508 | BMW M<br>Series M4<br>Coupe                           | 64.90 Lakh          | 29,000 kms | Petrol    | Automatic    | 2nd |
| 5509 | Jaguar XF<br>2.2 Litre<br>Luxury                      | 13.75 Lakh          | 90,000 kms | Diesel    | Automatic    | 2nd |
| 5510 | BMW 7<br>Series<br>730Ld                              | 29.90 Lakh          | 79,000 kms | Diesel    | Automatic    | 3rd |
| 5511 | BMW 5<br>Series<br>520d M<br>Sport                    | 31.90 Lakh          | 42,000 kms | Diesel    | Automatic    | 2nd |

5500 rows × 10 columns

```
In [28]: df["kms_driven"] = df["kms_driven"].str.replace(" kms", "")  
df["kms_driven"] = df["kms_driven"].str.replace(", ", "")
```

```
In [31]: df["kms_driven"] = df["kms_driven"].astype("int")
```

```
In [32]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 5500 entries, 0 to 5511
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   car_name               5500 non-null   object
1   car_prices_in_rupee    5500 non-null   object
2   kms_driven             5500 non-null   int64
3   fuel_type              5500 non-null   object
4   transmission           5500 non-null   object
5   ownership              5500 non-null   object
6   manufacture            5500 non-null   int64
7   engine                 5500 non-null   object
8   Seats                  5500 non-null   object
9   price is Lakh          5500 non-null   float64
dtypes: float64(1), int64(2), object(7)
memory usage: 472.7+ KB

```

```
In [33]: df["fuel_type"].unique()
```

```
Out[33]: array(['Diesel', 'Petrol', 'Cng', 'Electric', 'Lpg'], dtype=object)
```

```
In [34]: func1= lambda x: True if x=="Diesel" else False
func2= lambda x: True if x=="Petrol" else False
func3= lambda x: True if x=="Cng" else False
func4= lambda x: True if x=="Electric" else False
func5= lambda x: True if x=="Lpg" else False
```

```
In [37]: df["is Diesel"]= df["fuel_type"].apply(func1)
df["is Petrol"]= df["fuel_type"].apply(func2)
df["is Cng"]= df["fuel_type"].apply(func3)
df["is Electric"]= df["fuel_type"].apply(func4)
df["is Lpg"]= df["fuel_type"].apply(func5)
```

```
In [38]: df
```

Out[38]:

|      | car_name                                              | car_prices_in_rupee | kms_driven | fuel_type | transmission | own |
|------|-------------------------------------------------------|---------------------|------------|-----------|--------------|-----|
| 0    | Jeep<br>Compass<br>2.0<br>Longitude<br>Option<br>BSIV | 10.03 Lakh          | 86226      | Diesel    | Manual       | 1st |
| 1    | Renault<br>Duster RXZ<br>Turbo CVT                    | 12.83 Lakh          | 13248      | Petrol    | Automatic    | 1st |
| 2    | Toyota<br>Camry 2.5<br>G                              | 16.40 Lakh          | 60343      | Petrol    | Automatic    | 1st |
| 3    | Honda Jazz<br>VX CVT                                  | 7.77 Lakh           | 26696      | Petrol    | Automatic    | 1st |
| 5    | Volkswagen                                            | 7.66 Lakh           | 49719      | Petrol    | Automatic    | 1st |
| ...  | ...                                                   | ...                 | ...        | ...       | ...          | ... |
| 5507 | BMW X1<br>sDrive 20d<br>xLine                         | 28.90 Lakh          | 45000      | Diesel    | Automatic    | 1st |
| 5508 | BMW M<br>Series M4<br>Coupe                           | 64.90 Lakh          | 29000      | Petrol    | Automatic    | 2nd |
| 5509 | Jaguar XF<br>2.2 Litre<br>Luxury                      | 13.75 Lakh          | 90000      | Diesel    | Automatic    | 2nd |
| 5510 | BMW 7<br>Series<br>730Ld                              | 29.90 Lakh          | 79000      | Diesel    | Automatic    | 3rd |
| 5511 | BMW 5<br>Series<br>520d M<br>Sport                    | 31.90 Lakh          | 42000      | Diesel    | Automatic    | 2nd |

5500 rows × 15 columns

```
In [39]: df["transmission"].unique()
```

```
Out[39]: array(['Manual', 'Automatic'], dtype=object)
```

```
In [40]: func6= lambda x: True if x=="Manual" else False
```

```
In [41]: df["is Manual"]= df["transmission"].apply(func6)
```

```
In [42]: df
```

```
Out[42]:
```

|             | car_name                                              | car_prices_in_rupee | kms_driven | fuel_type | transmission | own |
|-------------|-------------------------------------------------------|---------------------|------------|-----------|--------------|-----|
| <b>0</b>    | Jeep<br>Compass<br>2.0<br>Longitude<br>Option<br>BSIV | 10.03 Lakh          | 86226      | Diesel    | Manual       | 1st |
| <b>1</b>    | Renault<br>Duster RXZ<br>Turbo CVT                    | 12.83 Lakh          | 13248      | Petrol    | Automatic    | 1st |
| <b>2</b>    | Toyota<br>Camry 2.5<br>G                              | 16.40 Lakh          | 60343      | Petrol    | Automatic    | 1st |
| <b>3</b>    | Honda Jazz<br>VX CVT                                  | 7.77 Lakh           | 26696      | Petrol    | Automatic    | 1st |
| <b>5</b>    | Volkswagen                                            | 7.66 Lakh           | 49719      | Petrol    | Automatic    | 1st |
| ...         | ...                                                   | ...                 | ...        | ...       | ...          | ... |
| <b>5507</b> | BMW X1<br>sDrive 20d<br>xLine                         | 28.90 Lakh          | 45000      | Diesel    | Automatic    | 1st |
| <b>5508</b> | BMW M<br>Series M4<br>Coupe                           | 64.90 Lakh          | 29000      | Petrol    | Automatic    | 2nd |
| <b>5509</b> | Jaguar XF<br>2.2 Litre<br>Luxury                      | 13.75 Lakh          | 90000      | Diesel    | Automatic    | 2nd |
| <b>5510</b> | BMW 7<br>Series<br>730Ld                              | 29.90 Lakh          | 79000      | Diesel    | Automatic    | 3rd |
| <b>5511</b> | BMW 5<br>Series<br>520d M<br>Sport                    | 31.90 Lakh          | 42000      | Diesel    | Automatic    | 2nd |

5500 rows × 16 columns

```
In [43]: df2= df
```

```
In [47]: df2.drop(["fuel_type","transmission","car_prices_in_rupee"], axis= 1, inplace=
```

```
In [48]: df2
```

Out[48]:

|      | car_name                               | kms_driven | ownership | manufacture | engine  | Seats   | price is Lakh | D |
|------|----------------------------------------|------------|-----------|-------------|---------|---------|---------------|---|
| 0    | Jeep Compass 2.0 Longitude Option BSIV | 86226      | 1st Owner | 2017        | 1956 cc | 5 Seats | 10.03         |   |
| 1    | Renault Duster RXZ Turbo CVT           | 13248      | 1st Owner | 2021        | 1330 cc | 5 Seats | 12.83         |   |
| 2    | Toyota Camry 2.5 G                     | 60343      | 1st Owner | 2016        | 2494 cc | 5 Seats | 16.40         |   |
| 3    | Honda Jazz VX CVT                      | 26696      | 1st Owner | 2018        | 1199 cc | 5 Seats | 7.77          |   |
| 5    | Volkswagen                             | 49719      | 1st Owner | 2017        | 1197 cc | 5 Seats | 7.66          |   |
| ...  | ...                                    | ...        | ...       | ...         | ...     | ...     | ...           |   |
| 5507 | BMW X1 sDrive 20d xLine                | 45000      | 1st Owner | 2018        | 2995 cc | 7 Seats | 28.90         |   |
| 5508 | BMW M Series M4 Coupe                  | 29000      | 2nd Owner | 2015        | 1968 cc | 5 Seats | 64.90         |   |
| 5509 | Jaguar XF 2.2 Litre Luxury             | 90000      | 2nd Owner | 2013        | 2755 cc | 5 Seats | 13.75         |   |
| 5510 | BMW 7 Series 730Ld                     | 79000      | 3rd Owner | 2015        | 2967 cc | 6 Seats | 29.90         |   |
| 5511 | BMW 5 Series 520d M Sport              | 42000      | 2nd Owner | 2017        | 1991 cc | 5 Seats | 31.90         |   |

5500 rows × 13 columns

In [ ]: